

Reasoning-Heavy Paper

Disclaimer: This is a pattern-based predicted practice paper workflow output, not a leaked, exact, or official CAT paper.

Variant Blueprint

- Variant type: reasoning_heavy
- Strategy: Favor hard/medium reasoning depth.
- Target section mix: QA 22 from the frozen selected pool; VARC/DILR are outside the current candidate scope
- Expected overlap score: 0.858
- Risk score: 0.35
- Diversity score: 0.727

Questions

Q1. QA - Arithmetic / Profit and Loss

- Archetype/type: Profit-loss transaction problem
- Difficulty: Hard
- Score: 0.91

A shop buys 90 identical bags at Rs 640 each. It marks every bag at the same price. Of the 90 bags, 36 are sold at a 10% discount, 30 at a 20% discount, 12 at a 35% discount because of minor defects, and 12 remain unsold. If the shop still makes an overall profit of Rs 6600 on the entire purchase, what is the marked price of each bag?

A. Rs 900 B. Rs 960 C. Rs 1000 D. Rs 1080

Q2. QA - Arithmetic / Mixtures and Alligation

- Archetype/type: Mixture concentration problem
- Difficulty: Hard
- Score: 0.905

Solution A is 40% acid by volume and Solution B is 70% acid by volume. A chemist mixes a litres of Solution A with b litres of Solution B. After mixing, exactly 10 litres of pure water evaporates from the combined mixture. The remaining mixture has volume 40 litres and is 75% acid. Find the ratio $a : b$.

A. 1 : 1 B. 1 : 2 C. 2 : 3 D. 3 : 5

Q3. QA - Arithmetic / Percentages

- Archetype/type: Successive percentage change
- Difficulty: Hard
- Score: 0.892

A value is first increased by $p\%$, then decreased by 20% , and then again increased by $p\%$. The final value is exactly 90% of what the value would have been after only a single $p\%$ increase from the original. If p is positive, find p .

Q4. QA - Arithmetic / Ratio and Proportion

- Archetype/type: Ratio sharing problem
- Difficulty: Hard
- Score: 0.873

The monthly incomes of P and Q are in the ratio 8:11, while their monthly expenditures are in the ratio 5:7. Q saves Rs 3000 more than P. After P invests 20% of his saving and Q invests 25% of his saving, their remaining savings are equal. What is Q's monthly income in rupees?

Q5. QA - Geometry / Triangles

- Archetype/type: Triangle similarity problem
- Difficulty: Medium-Hard
- Score: 0.865

In triangle ABC , point D lies on side AB and point E lies on side AC such that $DE \parallel BC$. The line through D parallel to AC meets BC at point F . If the area of triangle ADE is 16 and the area of quadrilateral $DECF$ is 40, find the area of triangle DBF .

A. 20 B. 24 C. 25 D. 36

Q6. QA - Arithmetic / Time and Work

- Archetype/type: Work-efficiency problem
- Difficulty: Hard
- Score: 0.862

A and B together can finish a job in 16 days, B and C together in 24 days, and A and C together in 18 days. A starts alone. After 4 days, C joins. After 3 more days, B also joins. How many days from the start are needed to finish the job?

A. 15.25 days B. $357/23$ days C. 16 days D. $371/24$ days

Q7. QA - Arithmetic / Averages

- Archetype/type: Weighted average problem
- Difficulty: Hard
- Score: 0.859

In a batch of 40 candidates, the average score is 69.6. The highest 5 candidates have an average score of 92, and the lowest 7 candidates have an average score of 48. The lowest 7 candidates are replaced by 7 new candidates. If the new

average score of all 40 candidates becomes 74.5, find the average score of the 7 new candidates.

Q8. QA - Algebra / Inequalities

- Archetype/type: Inequality optimization problem
- Difficulty: Hard
- Score: 0.855

For integer k with $0 \leq k \leq 15$, how many values of k make both roots of

$$x^2 + (k - 3)x + (k - 2) = 0$$

distinct negative real numbers?

A. 7 B. 8 C. 9 D. 12

Q9. QA - Algebra / Linear Equations

- Archetype/type: Integer feasibility under hidden divisibility constraint
- Difficulty: Hard
- Score: 0.854

A company has three departments X, Y, and Z. Every employee belongs to exactly one department. The number of employees in X is a positive multiple of 4. The number of employees in Y exceeds that in X by a positive multiple of 3. The number of employees in Z equals exactly half the sum of the employees in X and Y. If the total number of employees in the company is between 100 and 130, inclusive, how many distinct values can the total number of employees take?

Q10. QA - Algebra / Functions

- Archetype/type: Function value/domain problem
- Difficulty: Hard
- Score: 0.852

A function f defined on positive integers satisfies $f(1) = 1$ and

$$f(n) = n - f(f(n - 1)) \quad \text{for } n \geq 2.$$

Find $f(20)$.

A. 10 B. 11 C. 12 D. 13

Q11. QA - Algebra / Logarithms

- Archetype/type: Base-change identity with algebraic cancellation
- Difficulty: Hard
- Score: 0.842

Let $a = \log_{12} 18$ and $b = \log_{24} 54$. ; What is the value of

$$\frac{1 - ab}{a - b} ?$$

A. 3 B. 5 C. 7 D. -5

Q12. QA - Geometry / Coordinate Geometry

- Archetype/type: Coordinate geometry line/slope/distance problem
- Difficulty: Hard
- Score: 0.839

In a parallelogram $ABCD$ with vertices taken in that order, $A = (1, 2)$ and $B = (7, 5)$. The vertex $C = (p, q)$ has integer coordinates and $p > 10$. The diagonals intersect at a point lying on the line $x + 2y = 16$, and the area of the parallelogram is 42. Find $p + q$.

Q13. QA - Arithmetic / Time Speed Distance

- Archetype/type: Relative speed problem
- Difficulty: Hard
- Score: 0.83

Train A crosses a 150 m platform in 20 seconds. It overtakes a 180 m long train B, moving in the same direction, in 45 seconds. If train A is 250 m long, what is the speed of train B?

A. 32.4 km/h B. 36 km/h C. 37.6 km/h D. 40 km/h

Q14. QA - Algebra / Progressions

- Archetype/type: AP/GP sequence problem
- Difficulty: Hard
- Score: 0.824

An arithmetic progression has positive integer first term a and positive integer common difference d . The sum of the first 7 terms equals the sum of the next 5 terms, i.e. the sum of the 8th to 12th terms. If the 10th term lies strictly between 50 and 80, how many ordered pairs (a, d) are possible?

A. 1 B. 2 C. 3 D. 4

Q15. QA - Geometry / Mensuration

- Archetype/type: Mensuration volume/area problem
- Difficulty: Hard
- Score: 0.811

A right circular cone has base radius 10 cm and height 24 cm. It is cut by a plane parallel to the base so that the curved surface area of the smaller cone removed from the top is one-fourth of the curved surface area of the original cone. The remaining frustum is melted and recast into identical solid cylinders, each of radius 5 cm and height 7 cm. How many such cylinders can be made?

A. 3 B. 4 C. 5 D. 6

Q16. QA - Number System / Divisibility

- Archetype/type: Divisibility/factor-counting problem
- Difficulty: Hard
- Score: 0.808

Find the least positive integer n such that n leaves remainders 3, 5, and 7 when divided by 5, 7, and 9, respectively, and n is divisible by 11.

A. 943 B. 1258 C. 1573 D. 1888

Q17. QA - Arithmetic / Percentages

- Archetype/type: Successive percentage change
- Difficulty: Hard
- Score: 0.887

In January, revenue from product X was exactly 40% of a firm's total revenue. In February, total revenue increased by 25%, but revenue from X decreased by 10% because only some contracts were renewed. By what percentage did revenue from all other products increase from January to February?

Q18. QA - Arithmetic / Percentages

- Archetype/type: Successive percentage change
- Difficulty: Medium-Hard
- Score: 0.872

In an entrance test, 80% of candidates clear Section A. Of those who clear Section A, 75% clear Section B. Of those who clear both sections, 40% clear Section C. Among the candidates who do not clear all three sections, the candidates who failed only Section C (i.e., cleared A and B but failed C) form $\frac{9}{19}$ of that group. If the number of such candidates is 720, find the total number of candidates who appeared.

Q19. QA - Arithmetic / Percentages

- Archetype/type: Successive percentage change
- Difficulty: Hard
- Score: 0.872

In an entrance test, 40% of the candidates are women. Exactly 30% of the men and 45% of the women qualify. Later, no qualified man withdraws, but some qualified women withdraw. After withdrawals, qualified men and qualified women are in the ratio 3:2. What percentage of the qualified women withdrew?

Q20. QA - Arithmetic / Ratio and Proportion

- Archetype/type: Ratio sharing problem
- Difficulty: Hard
- Score: 0.87

A fund is divided among A, B and C in the ratio 5:7:8. B gives Rs 22000 from his share to A, and then C gives one-fourth of his new comparison excess over A to B. After both transfers, A and B have equal amounts. What was C's original share in rupees?

Q21. QA - Algebra / Functions

- Archetype/type: Function value/domain problem
- Difficulty: Hard
- Score: 0.852

For real x , define

$$f(x) = \begin{cases} \frac{x-2}{x+1}, & x < 1, \ x \neq -1, \\ \sqrt{x-1}, & x \geq 1. \end{cases}$$

For how many integers x in the interval $[-10, 20]$ is $f(f(x))$ defined and greater than 1?

A. 14 B. 15 C. 16 D. 17

Q22. QA - Arithmetic / Ratio and Proportion

- Archetype/type: Ratio sharing problem
- Difficulty: Hard
- Score: 0.843

Two boxes A and B contain tokens in the ratio 13:8. Some tokens are moved from A to B, after which the ratio becomes 5:4. If 42 more tokens are then moved from A to B and the two boxes become equal, what was the total number of tokens initially?

Answer Key

Q	Answer	Score
1	Rs 1000	0.91
2	(B) 1 : 2	0.905

Q	Answer	Score
3	12.5	0.892
4	825000	0.873
5	(C) 25	0.865
6	357/23 days	0.862
7	76	0.859
8	(B) 8	0.855
9	10	0.854
10	(C) 12	0.852
11	(B) 5	0.842
12	23	0.839
13	37.6 km/h	0.83
14	(A) 1	0.824
15	(B) 4	0.811
16	(C) 1573	0.808
17	145/3	0.887
18	2000	0.872
19	100/3	0.872
20	144000	0.87
21	(C) 16	0.852
22	756	0.843

Detailed Solutions

Q1

Step 1: Total cost of all 90 bags = $90 \times 640 = \text{Rs } 57600$. Step 2: Profit is calculated on the entire purchase, including the unsold bags, so required revenue = $57600 + 6600 = \text{Rs } 64200$. Step 3: Let the marked price be M . Only 78 bags are sold, giving revenue $36 \times 0.90M + 30 \times 0.80M + 12 \times 0.65M = 32.4M + 24M + 7.8M = 64.2M$. Step 4: Therefore $64.2M = 64200$, so $M = \text{Rs } 1000$. Step 5: Verification: revenue = $36 \times 900 + 30 \times 800 + 12 \times 650 = 32400 + 24000 + 7800 = \text{Rs } 64200$.

Q2

Step 1: After evaporation, the remaining volume is 40 litres and the acid concentration is 75%. Hence acid in the final mixture is

$$0.75 \times 40 = 30 \text{ litres.}$$

Step 2: Evaporation removes only water, not acid. Therefore acid before evaporation was also 30 litres. Step 3: Since 10 litres of water evaporated and final volume is 40 litres, the pre-evaporation mixed volume was

$$40 + 10 = 50 \text{ litres.}$$

So

$$a + b = 50.$$

Step 4: Acid balance before evaporation:

$$0.40a + 0.70b = 30.$$

Step 5: Substitute $a = 50 - b$:

$$0.40(50 - b) + 0.70b = 30,$$

$$20 - 0.40b + 0.70b = 30,$$

$$0.30b = 10 \Rightarrow b = \frac{100}{3}.$$

Then

$$a = 50 - \frac{100}{3} = \frac{50}{3}.$$

Step 6: Therefore

$$a : b = \frac{50}{3} : \frac{100}{3} = 1 : 2.$$

Q3

Step 1: Let the original value be 100. Step 2: After the three-stage change, the value is $100(1+p/100)(0.8)(1+p/100)$. Step 3: After only a single $p\%$ increase, the value would be $100(1+p/100)$. Step 4: The first value is 90% of the second, so $0.8(1+p/100)^2 = 0.9(1+p/100)$. Step 5: Since p is positive, divide by the common positive factor to get $0.8(1+p/100)=0.9$, so $p=12.5$.

Q4

Step 1: Let incomes be $8x$ and $11x$, and expenditures be $5y$ and $7y$. Then P's saving is $8x - 5y$ and Q's saving is $11x - 7y$. Step 2: Q saves Rs 3000 more, so $11x - 7y = 8x - 5y + 3000$, giving $3x - 2y = 3000$. Step 3: After investment, remaining savings are 80% of P's saving and 75% of Q's saving, so $0.8(8x - 5y) = 0.75(11x - 7y)$. Since Q's saving exceeds P's by 3000, this is also $0.8(SQ - 3000) = 0.75SQ$. Hence $SQ = 48000$. Step 4: Therefore $11x - 7y = 48000$ and $3x - 2y = 3000$. From the second equation, $y = 1.5x - 1500$. Step 5: Substitute: $11x - 7(1.5x - 1500) = 48000$, so $0.5x + 10500 = 48000$, $x = 75000$. Q's income = $11x = \text{Rs } 825000$.

Q5

Step 1: Let $\frac{AD}{AB} = k$. Since $DE \parallel BC$, triangle $ADE \sim ABC$ with linear ratio k , so $[ADE] = k^2 \cdot [ABC]$. Step 2: Since $DF \parallel AC$, triangle $DBF \sim ABC$ (same angle at B , with $DB/AB = 1 - k$ and $BF/BC = 1 - k$). Hence $[DBF] = (1 - k)^2 \cdot [ABC]$. Step 3: Identify the hidden structure of $DECF$.

Both $DE \parallel BC$ and $DF \parallel AC$. By the proportionality theorem, $DE = k \cdot BC$ and $FC = BC - BF = (1 - (1 - k)) \cdot BC = k \cdot BC$. So $DE = FC$ with $DE \parallel FC$, meaning $DECF$ is a *parallelogram*, not just a trapezoid. Step 4: Since the three regions $[ADE]$, $[DECF]$, $[DBF]$ partition $[ABC]$:

$$[ABC] = [ADE] + [DBF] + [DECF] = k^2T + (1 - k)^2T + 2k(1 - k)T = T.$$

So $[DECF] = 2k(1 - k)T$. Step 5: Now $[DECF]^2 = 4k^2(1 - k)^2T^2 = 4 \cdot [ADE] \cdot [DBF]$. Substituting:

$$40^2 = 4 \cdot 16 \cdot [DBF] \Rightarrow [DBF] = \frac{1600}{64} = 25.$$

Q6

Let the daily rates of A, B and C be a , b and c . We have $a + b = 1/16$, $b + c = 1/24$, and $a + c = 1/18$. Adding gives $2(a + b + c) = 1/16 + 1/24 + 1/18 = 9/144 + 6/144 + 8/144 = 23/144$, so $a + b + c = 23/288$. Also $a = [(a + b) + (a + c) - (b + c)]/2 = (1/16 + 1/18 - 1/24)/2 = (9 + 8 - 6)/144$ divided by 2 = $11/288$. Similarly $c = [(b + c) + (a + c) - (a + b)]/2 = (6 + 8 - 9)/144$ divided by 2 = $5/288$. In the first 4 days A completes $4 \times 11/288 = 44/288 = 11/72$. In the next 3 days A + C complete $3 \times (16/288) = 48/288 = 1/6$. Work done in first 7 days = $11/72 + 1/6 = 23/72$. Remaining work = $49/72$. With all three working, time needed = $(49/72)/(23/288) = 196/23$ days. Total time = $7 + 196/23 = 357/23$ days.

Q7

Step 1: Convert the original average into the original total:

$$40 \times 69.6 = 2784.$$

Step 2: The total score of the lowest 7 candidates is

$$7 \times 48 = 336.$$

Step 3: After replacing these 7 candidates, the total score of the unchanged 33 candidates is

$$2784 - 336 = 2448.$$

Step 4: The new average of all 40 candidates is 74.5, so the new total score is

$$40 \times 74.5 = 2980.$$

Step 5: Therefore, the total score of the 7 new candidates is

$$2980 - 2448 = 532.$$

Step 6: Hence their average score is

$$\frac{532}{7} = 76.$$

Q8

Step 1: For both roots negative and *distinct* real:

- Discriminant > 0 (strict, for distinct): $(k-3)^2 - 4(k-2) > 0$.
- Sum of roots negative: $-(k-3) < 0$, i.e. $k > 3$.
- Product of roots positive: $k-2 > 0$, i.e. $k > 2$.

Step 2: Simplify discriminant: $(k-3)^2 - 4(k-2) = k^2 - 6k + 9 - 4k + 8 = k^2 - 10k + 17 > 0$. Step 3: Roots of $k^2 - 10k + 17 = 0$: $k = \frac{10 \pm \sqrt{100 - 68}}{2} = \frac{10 \pm \sqrt{32}}{2} = 5 \pm 2\sqrt{2}$. Approximately $5 + 2\sqrt{2} \approx 7.83$ and $5 - 2\sqrt{2} \approx 2.17$.

Step 4: So $k^2 - 10k + 17 > 0$ iff $k < 5 - 2\sqrt{2}$ or $k > 5 + 2\sqrt{2}$. Step 5: Combine with $k > 3$ and $k > 2$: intersection is $k > 5 + 2\sqrt{2} \approx 7.83$ (the branch $k < 5 - 2\sqrt{2} \approx 2.17$ is killed by $k > 3$). Step 6: Integer k with $7.83 < k \leq 15$: $k \in \{8, 9, 10, 11, 12, 13, 14, 15\}$. Count = 8. Step 7: Verify endpoints. $k = 8$: $x^2 + 5x + 6 = (x+2)(x+3) = 0$, roots $-2, -3$, both negative and distinct $\checkmark$. $k = 7$: $x^2 + 4x + 5 = 0$, discriminant $= 16 - 20 = -4 < 0$, complex roots $\times$ (good).

Q9

Step 1: Let the number of employees in X be $4a$ where $a \geq 1$, and let Y exceed X by $3b$ where $b \geq 1$. Then $|Y| = 4a + 3b$. Step 2: The number of employees in Z is

$$|Z| = \frac{|X| + |Y|}{2} = \frac{4a + 4a + 3b}{2} = \frac{8a + 3b}{2}.$$

For $|Z|$ to be a positive integer, $8a + 3b$ must be even. Since $8a$ is always even, $3b$ must be even, which forces b to be even. Write $b = 2k$ with $k \geq 1$. Step 3: With $b = 2k$, we get $|Z| = 4a + 3k$ and

$$T = |X| + |Y| + |Z| = 4a + (4a + 6k) + (4a + 3k) = 12a + 9k = 3(4a + 3k).$$

So T is always a multiple of 3. Step 4: Impose $100 \leq T \leq 130$:

$$100 \leq 3(4a + 3k) \leq 130 \implies 34 \leq 4a + 3k \leq 43.$$

Let $m = 4a + 3k$ where $a \geq 1$ and $k \geq 1$, so $m \geq 7$. For any integer $m \geq 7$, a representation $m = 4a + 3k$ with $a, k \geq 1$ exists: if $m \equiv 1 \pmod{3}$, take $a = 1$; if $m \equiv 2 \pmod{3}$, take $a = 2$; if $m \equiv 0 \pmod{3}$, take $a = 3$; and solve for k accordingly. Each resulting k is positive for $m \geq 7$. Step 5: The integers m from 34 to 43 inclusive give $43 - 34 + 1 = 10$ values of m , hence 10 distinct totals $T \in \{102, 105, 108, 111, 114, 117, 120, 123, 126, 129\}$.

Q10

Step 1: Compute $f(n)$ step by step. $f(1) = 1.$ \ $f(2) = 2 - f(f(1)) = 2 - f(1) = 1.$ \ $f(3) = 3 - f(f(2)) = 3 - f(1) = 2.$ \ $f(4) = 4 - f(f(3)) = 4 - f(2) = 3.$ \ $f(5) = 5 - f(f(4)) = 5 - f(3) = 3.$ \ $f(6) = 6 - f(f(5)) = 6 - f(3) = 4.$ \ $f(7) = 7 - f(f(6)) = 7 - f(4) = 4.$ \ $f(8) = 8 - f(f(7)) = 8 - f(4) = 5.$
 Step 2: Continue. $f(9) = 9 - f(f(8)) = 9 - f(5) = 6.$ \ $f(10) = 10 - f(f(9)) = 10 - f(6) = 6.$ \ $f(11) = 11 - f(f(10)) = 11 - f(6) = 7.$ \ $f(12) = 12 - f(f(11)) = 12 - f(7) = 8.$ Step 3: Continue. $f(13) = 13 - f(f(12)) = 13 - f(8) = 8.$ \ $f(14) = 14 - f(f(13)) = 14 - f(8) = 9.$ \ $f(15) = 15 - f(f(14)) = 15 - f(9) = 9.$ \ $f(16) = 16 - f(f(15)) = 16 - f(9) = 10.$ Step 4: Continue. $f(17) = 17 - f(f(16)) = 17 - f(10) = 11.$ \ $f(18) = 18 - f(f(17)) = 18 - f(11) = 11.$ \ $f(19) = 19 - f(f(18)) = 19 - f(11) = 12.$ \ $f(20) = 20 - f(f(19)) = 20 - f(12) = 20 - 8 = 12.$ Step 5: $f(20) = 12.$

Q11

Step 1: Set $p = \ln 2$ and $q = \ln 3$. Then

$$a = \frac{\ln 18}{\ln 12} = \frac{p + 2q}{2p + q}, \quad b = \frac{\ln 54}{\ln 24} = \frac{p + 3q}{3p + q}.$$

Step 2: Compute $a - b$ over the common denominator $(2p + q)(3p + q)$:

$$a - b = \frac{(p + 2q)(3p + q) - (p + 3q)(2p + q)}{(2p + q)(3p + q)}.$$

Expanding the numerator:

$$(3p^2 + 7pq + 2q^2) - (2p^2 + 7pq + 3q^2) = p^2 - q^2 = (p - q)(p + q).$$

Step 3: Compute ab :

$$ab = \frac{(p + 2q)(p + 3q)}{(2p + q)(3p + q)} = \frac{p^2 + 5pq + 6q^2}{6p^2 + 5pq + q^2}.$$

Step 4: Compute $1 - ab$:

$$1 - ab = \frac{(6p^2 + 5pq + q^2) - (p^2 + 5pq + 6q^2)}{(2p + q)(3p + q)} = \frac{5p^2 - 5q^2}{(2p + q)(3p + q)} = \frac{5(p - q)(p + q)}{(2p + q)(3p + q)}.$$

Step 5: Form the ratio:

$$\frac{1 - ab}{a - b} = \frac{5(p - q)(p + q)}{(2p + q)(3p + q)} \cdot \frac{(2p + q)(3p + q)}{(p - q)(p + q)} = 5.$$

Note that $p \neq q$ (since $\ln 2 \neq \ln 3$) and $p + q \neq 0$, so the cancellation is valid.

Q12

Step 1: In a parallelogram, the diagonals bisect each other. Hence the midpoint of AC is

$$\left(\frac{1+p}{2}, \frac{2+q}{2} \right).$$

Step 2: This point lies on $x + 2y = 16$, so

$$\frac{1+p}{2} + 2 \left(\frac{2+q}{2} \right) = 16.$$

Step 3: Simplify:

$$\frac{1+p}{2} + 2 + q = 16.$$

Multiplying by 2,

$$1 + p + 4 + 2q = 32,$$

so

$$p + 2q = 27.$$

Step 4: The adjacent side vectors are

$$\overrightarrow{AB} = (6, 3), \quad \overrightarrow{BC} = (p-7, q-5).$$

The area of the parallelogram is the absolute value of their determinant:

$$|6(q-5) - 3(p-7)| = 42.$$

Step 5: Substitute $p = 27 - 2q$:

$$|6q - 30 - 3(27 - 2q - 7)| = 42.$$

This becomes

$$|6q - 30 - 3(20 - 2q)| = 42,$$

or

$$|12q - 90| = 42.$$

Step 6: Therefore,

$$12q - 90 = 42 \quad \text{or} \quad 12q - 90 = -42.$$

So

$$q = 11 \quad \text{or} \quad q = 4.$$

Step 7: From $p + 2q = 27$, these give

$$(q, p) = (11, 5) \quad \text{or} \quad (4, 19).$$

The condition $p > 10$ selects

$$p = 19, \quad q = 4.$$

Hence

$$p + q = 23.$$

Q13

Step 1: While crossing the platform, Train A covers its own length plus platform length = $250 + 150 = 400$ m in 20 seconds. So speed of A = 20 m/s. Step 2: While overtaking Train B in the same direction, Train A must cover the sum of both train lengths relative to B: $250 + 180 = 430$ m. Step 3: Relative speed = $430/45 = 86/9$ m/s. Step 4: Speed of B = $20 - 86/9 = 94/9$ m/s. Step 5: Convert to km/h: $(94/9) \times 18/5 = 188/5 = 37.6$ km/h.

Q14

Step 1: Sum of the first 7 terms is

$$S_7 = \frac{7}{2}(2a + 6d) = 7a + 21d.$$

Step 2: Sum of the 8th to 12th terms is

$$(a + 7d) + (a + 8d) + (a + 9d) + (a + 10d) + (a + 11d) = 5a + 45d.$$

Step 3: Equating the two sums,

$$7a + 21d = 5a + 45d \Rightarrow 2a = 24d \Rightarrow a = 12d.$$

Step 4: The 10th term is

$$a + 9d = 12d + 9d = 21d.$$

The condition $50 < 21d < 80$ gives

$$\frac{50}{21} < d < \frac{80}{21}.$$

Step 5: Since d is a positive integer, the only possible value is $d = 3$. Then $a = 36$. Hence the only ordered pair is $(36, 3)$.

Q15

Step 1: The original cone has radius 10 and height 24, so its slant height is

$$\sqrt{10^2 + 24^2} = 26.$$

Step 2: The curved surface area of a cone is proportional to rl . For similar cones, both r and l scale by the same linear factor, so curved surface area scales as the square of the linear factor. Step 3: Since the smaller cone has one-fourth the curved surface area of the original cone, its linear scale factor is

$$\sqrt{\frac{1}{4}} = \frac{1}{2}.$$

Step 4: Therefore the volume of the smaller cone is

$$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

of the original cone's volume. Step 5: The original cone's volume is

$$\frac{1}{3}\pi(10)^2(24) = 800\pi.$$

So the frustum volume is

$$800\pi - \frac{1}{8}(800\pi) = 700\pi.$$

Step 6: Each cylinder has volume

$$\pi(5)^2(7) = 175\pi.$$

Therefore the number of cylinders is

$$\frac{700\pi}{175\pi} = 4.$$

Q16

Step 1: The given remainder conditions are

$$n \equiv 3 \pmod{5}, \quad n \equiv 5 \pmod{7}, \quad n \equiv 7 \pmod{9}.$$

Step 2: Each remainder is 2 less than the corresponding divisor. Hence

$$n + 2$$

is divisible by 5, 7, and 9. Step 3: Since

$$\text{lcm}(5, 7, 9) = 315,$$

we can write

$$n + 2 = 315k,$$

so

$$n = 315k - 2.$$

Step 4: Now impose divisibility by 11:

$$315k - 2 \equiv 0 \pmod{11}.$$

Step 5: Since

$$315 \equiv 7 \pmod{11},$$

we get

$$7k - 2 \equiv 0 \pmod{11},$$

or

$$7k \equiv 2 \pmod{11}.$$

Step 6: The inverse of 7 modulo 11 is 8, because

$$7 \cdot 8 = 56 \equiv 1 \pmod{11}.$$

Therefore,

$$k \equiv 2 \cdot 8 = 16 \equiv 5 \pmod{11}.$$

Step 7: The least positive value is $k = 5$. Hence

$$n = 315(5) - 2 = 1575 - 2 = 1573.$$

Q17

Step 1: Let January total revenue be 100, so X revenue is 40 and other revenue is 60. Step 2: February total revenue is 125 after the 25% increase. Step 3: February X revenue is 90% of 40, which is 36. Step 4: Therefore February revenue from other products is $125 - 36 = 89$. Step 5: The increase in other-products revenue is 29 on a base of 60, so the percentage increase is $2900/60 = 145/3\%$.

Q18

Step 1: Let total candidates = N . Clear A = $0.80N$. Clear A and B = $0.75 \times 0.80N = 0.60N$. Step 2: Clear all three = $0.40 \times 0.60N = 0.24N$. Hence candidates who do not clear all three = $N - 0.24N = 0.76N$. Step 3: Failed only C means cleared A and B but failed C, so this group is $0.60N - 0.24N = 0.36N$. Step 4: The given fraction is consistent because $0.36N/0.76N = 36/76 = 9/19$. Step 5: Given $0.36N = 720$, we get $N = 720/0.36 = 2000$.

Q19

Step 1: Let the total number of candidates be 100. Step 2: There are 60 men and 40 women; qualified men are 30% of 60 = 18. Step 3: Qualified women are 45% of 40 = 18 before withdrawals. Step 4: After withdrawals, men:women is 3:2, so if men are 18, remaining women are 12. Step 5: Women who withdrew = $18 - 12 = 6$, which is $6/18 = 100/3\%$ of qualified women.

Q20

Step 1: Let the original shares be $5x$, $7x$ and $8x$. After B gives Rs 22000 to A, A has $5x + 22000$ and B has $7x - 22000$. C still has $8x$. Step 2: C's excess over A at this point is $8x - (5x + 22000) = 3x - 22000$. One-fourth of this amount is transferred from C to B, so B receives $(3x - 22000)/4$. Step 3: Final A = $5x + 22000$. Final B = $7x - 22000 + (3x - 22000)/4$. Step 4: They are equal, so $5x + 22000 = 7x - 22000 + (3x - 22000)/4$. Multiplying by 4 gives $20x + 88000 = 28x - 88000 + 3x - 22000 = 31x - 110000$. Step 5: Hence $11x = 198000$, so $x = 18000$. C's original share = $8x = \text{Rs } 144000$.

Q21

Step 1: The composition $f(f(x))$ is defined only if x is in the domain of f and the intermediate value $f(x)$ is also in the domain of f . Thus $x = -1$ is automatically excluded. Step 2: First consider $x < 1$. Then

$$f(x) = \frac{x-2}{x+1}.$$

For $f(f(x)) > 1$ with the intermediate input below 1, we need

$$\frac{y-2}{y+1} > 1, \quad \text{where } y = f(x).$$

This gives

$$\frac{-3}{y+1} > 0 \Rightarrow y < -1.$$

Step 3: Now solve

$$\frac{x-2}{x+1} < -1.$$

For $x < -1$, multiplying by the negative denominator reverses the sign:

$$x-2 > -x-1 \Rightarrow 2x > 1,$$

which gives no integer $x < -1$. For $-1 < x < 1$, the only integer is $x = 0$, and it gives

$$f(0) = -2, \quad f(-2) = 4 > 1.$$

So $x = 0$ works. Step 4: Now consider $x \geq 1$. Then

$$f(x) = \sqrt{x-1}.$$

If $0 \leq f(x) < 1$, then the first branch is used next and cannot exceed 1 because

$$\frac{y-2}{y+1} > 1$$

would require $y < -1$, which is impossible for $y \geq 0$. Step 5: Therefore we need $f(x) \geq 1$ and

$$f(f(x)) = \sqrt{\sqrt{x-1}-1} > 1.$$

This gives

$$\sqrt{x-1}-1 > 1 \Rightarrow \sqrt{x-1} > 2 \Rightarrow x > 5.$$

Step 6: The integers from 6 to 20 give 15 values. Including $x = 0$, the total number of valid integers is

$$15 + 1 = 16.$$

Q22

Step 1: Let the initial numbers be 13k and 8k. Suppose x tokens are first moved from A to B. Step 2: Then $(13k - x)/(8k + x) = 5/4$. Cross-multiplying gives $52k - 4x = 40k + 5x$, so $12k = 9x$ and $x = 4k/3$. Step 3: After this transfer, the boxes have $13k - 4k/3 = 35k/3$ and $8k + 4k/3 = 28k/3$ tokens. Step 4: Moving 42 more from A to B makes them equal, so $35k/3 - 42 = 28k/3 + 42$. Step 5: Therefore $7k/3 = 84$, so $k = 36$. Initial total = $21k = 756$.

Evidence Lineage

Q	Candidate	Source Question IDs
1	CAT_QA_SPEC_01_ECKME 2017	S1_QA_Q12;CAT_2017_S2_QA_Q07
2	CAT_QA_SPEC_10_CORRE2018	S10_QA_Q01;CAT_2018_S1_QA_Q14
3	CAT_QA_SPEC_06_ECKME2003	S1_QA_Q01;CAT_2017_S1_QA_Q05
4	CAT_QA_SPEC_03_ECKME 2017	S1_QA_Q13;CAT_2018_S1_QA_Q26
5	CAT_QA_SPEC_07_CCAUD2017	S1_QA_Q16;CAT_2018_S1_QA_Q09
6	CAT_QA_SPEC_02_CAND 2017	S1_QA_Q34;CAT_2017_S2_QA_Q34
7	CAT_QA_SPEC_15_GPAT5 2017	ki6g_QAG106;CAT_2017_S2_QA_Q26
8	CAT_QA_SPEC_08_CCAUD2017	S1_QA_Q03;CAT_2017_S1_QA_Q26
9	CAT_QA_SPEC_12_GENSPARK SHINQA	Q28;CAT_2018_S2_QA_Q29
10	CAT_QA_SPEC_13_CCAUD2018	S2_QA_Q30;CAT_2019_S1_QA_Q09
11	CAT_QA_SPEC_16_GENSPARK SHINQA	Q23;CAT_2018_S1_QA_Q11
12	CAT_QA_SPEC_18_GPAT5 2017	ki6g_QCOORDINENS 2019
13	CAT_QA_SPEC_04_ECKME 2017	S1_QA_Q04;CAT_2017_S2_QA_Q04
14	CAT_QA_SPEC_11_CORRE2017	S20_QA_Q15;CAT_2017_S2_QA_Q21
15	CAT_QA_SPEC_19_GPAT5 2017	ki6g_QCOORDINENS 2017
16	CAT_QA_SPEC_20_GPAT5 2019	ki6g_QAG106;CAT_2022_S2_QA_Q13
17	CAT_QA_SPEC_06_ECKME2007	S1_QA_Q01;CAT_2017_S1_QA_Q05
18	CAT_QA_SPEC_06_CCAUD2017	S2_QA_Q06;CAT_2017_S2_QA_Q10
19	CAT_QA_SPEC_06_ECKME2007	S1_QA_Q01;CAT_2017_S1_QA_Q05
20	CAT_QA_SPEC_03_ECKME 2018	S1_QA_Q26;CAT_2018_S2_QA_Q34
21	CAT_QA_SPEC_13_GPAT5 2017	ki6g_QUANCIRCAT_2017
22	CAT_QA_SPEC_03_ECKME 2017	S2_QA_Q14;CAT_2018_S1_QA_Q34